

The mutual positions of lines and planes in space

Every possible arrangement of two lines, of a line and a plane, and of two planes — nine cases in all.

LESSON 6–7

2 × 40 MIN

GEOMETRY 10, §4

IGCSE E4.7

LESSON CLOCK

8'

16'

20'

24'

14'

8'

Two lines — three cases	8 min
Line and plane; plane and plane	16 min
Finding skew pairs in a cuboid	20 min
The angle between skew lines	24 min
Practice	14 min
Homework	8 min

LEARNING OBJECTIVES

- List the three possible positions of two lines in space.
- List the three positions of a line and a plane, and the three of two planes.
- Recognise skew lines in a cuboid and name pairs of them.
- Find the angle between two skew lines by translating one of them.

TEXTBOOK

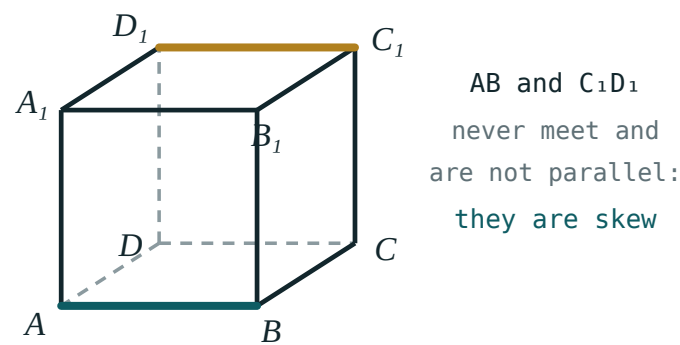
National	pp. 29–40
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 4.1

01

Explanation

Two lines: three cases, not two

CASE	COMMON POINTS	COPLANAR?
intersecting	1	yes
parallel	0	yes
skew	0	no



An edge of the top face and a non-touching edge of the bottom face. They never meet, and no plane holds both.

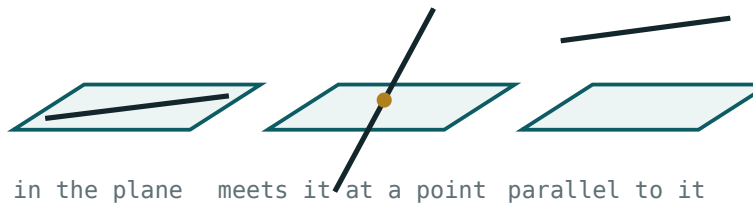
ZERO COMMON POINTS IS NOT ENOUGH

Parallel and skew lines both fail to meet. What separates them is whether a single plane can hold both. Always check for a common plane before writing “parallel”.

A line and a plane, and two planes

OBJECTS	CASE	COMMON POINTS
line and plane	line lies in the plane	infinitely many
	line meets the plane	1
	line parallel to the plane	0
two planes	coincident	infinitely many
	intersecting	a whole line
	parallel	0

a line and a plane



In the plane, meeting the plane at a point, and parallel to it.

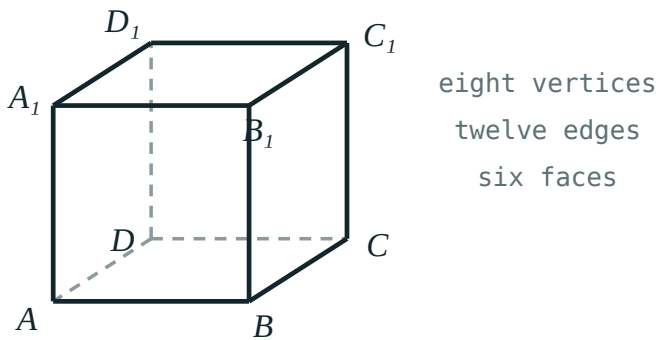
THE TEST FOR A LINE PARALLEL TO A PLANE

A line not in a plane is parallel to it exactly when it is parallel to **some** line lying in that plane. Finding that one line is the whole of most proofs.

Skew pairs in a cuboid

The cuboid $ABCD A_1 B_1 C_1 D_1$ is the standard laboratory. Take edge AB :

RELATION TO AB	EDGES
meets it	AD, BC, AA_1, BB_1
parallel to it	$DC, A_1 B_1, D_1 C_1$
skew to it	$CC_1, DD_1, A_1 D_1, B_1 C_1$



eight vertices
twelve edges
six faces

The standard labelling. Every question about a cube uses these eight letters.

Four skew, four meeting, three parallel — eleven other edges, and every one accounted for.

The angle between skew lines

THE METHOD

Skew lines never meet, so there is no angle to measure directly. **Translate one of them** until it passes through a point of the other; the angle between the resulting intersecting lines is defined to be the angle between the skew pair. It does not depend on the point chosen.

In a cube of side a , find the angle between AB and B_1C_1 . Translate B_1C_1 down to BC . Now $AB \perp BC$, so the answer is 90° — skew lines can be perpendicular without ever meeting.

angle between AB and DC_1 : translate DC_1 to AB_1 , then $\tan \theta = \frac{a}{a} = 1 \Rightarrow \theta = 45^\circ$

02 Worked examples

EXAMPLE 1 In cube $ABCD A_1 B_1 C_1 D_1$, name all edges skew to AA_1 .

- ① Edges meeting AA_1 : $AB, AD, A_1 B_1, A_1 D_1$.
- ② Edges parallel: BB_1, CC_1, DD_1 .
- ③ The rest are skew.

ANSWER

$BC, CD, B_1 C_1, C_1 D_1$

EXAMPLE 2 Cube of side a . Find the angle between AB_1 and BC_1 .

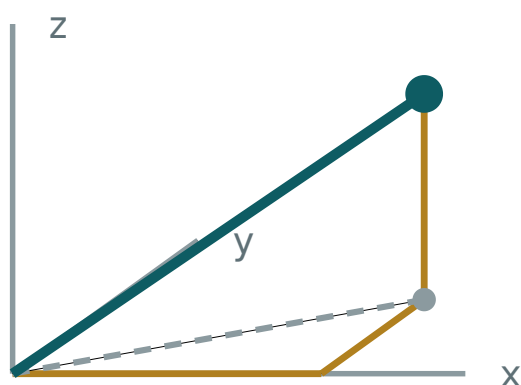
- ① Both are face diagonals, length $a\sqrt{2}$.
- ② Translate BC_1 to AD_1 — now they meet at A .
- ③ $B_1D_1 = a\sqrt{2}$
Triangle AB_1D_1 is equilateral.
- ④ All sides $a\sqrt{2}$.

ANSWER 60° **EXAMPLE 3** Two planes are each parallel to a line ℓ . Must they be parallel to each other?

- ① Consider two walls of a room and the vertical line where the floor meets nothing.
- ② Both can be parallel to a vertical line and still meet.

ANSWERNo — they may intersect in a line parallel to ℓ **03 Interactive model**

Hold two pencils as skew lines, then slide one until they meet, keeping its direction.

Lines in a box

x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to'g'ri chiziqlar	Скрещивающиеся прямые
Intersecting lines	Kesishuvchi to'g'ri chiziqlar	Пересекающиеся прямые
Parallel lines	Parallel to'g'ri chiziqlar	Параллельные прямые
Line parallel to a plane	Tekislikka parallel chiziq	Прямая, параллельная плоскости
Line in a plane	Tekislikdagi chiziq	Прямая, лежащая в плоскости
Intersecting planes	Kesishuvchi tekisliklar	Пересекающиеся плоскости
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Angle between skew lines	Ayqash chiziqlar orasidagi burchak	Угол между скрещивающимися прямыми
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Skew lines are:

parallel

intersecting

non-coplanar and non-intersecting

perpendicular

A line and a plane can share:

0, 1 or infinitely many points

only 0 or 1

only 1

exactly 2

Two distinct planes can share:

a point

a line

two points only

nothing but a point

The angle between skew lines is found by:

measuring the gap

translating one to meet the other

projecting onto the floor

it is undefined

In a cube, how many edges are skew to a given edge?

2

3

4

6

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the three positions of two lines in space

ANSWER intersecting, parallel, skew

2. Name the three positions of a line and a plane

ANSWER in it, meeting it, parallel to it

3. Name the three positions of two planes

ANSWER coincident, intersecting, parallel

4. In a cube, is AB parallel to D_1C_1 ?

ANSWER yes

5. In a cube, is AB skew to CC_1 ?

ANSWER yes

6. How many edges of a cube meet AB ?

ANSWER 4

7. Can skew lines be perpendicular?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. List all edges skew to BC in a cube

ANSWER $AA_1, DD_1, A_1B_1, C_1D_1$

2. Angle between AB and B_1C_1 in a cube

ANSWER 90°

3. Angle between AB and DC_1 in a cube

ANSWER 45°

4. Angle between AB_1 and BC_1 in a cube

ANSWER 60°

5. Two planes parallel to the same line — must they be parallel?

ANSWER no

6. Two lines parallel to the same plane — must they be parallel?

ANSWER no

7. Two planes parallel to the same plane — must they be parallel?

ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Cuboid $4 \times 3 \times 12$. Angle between AB and DB_1

ANSWER $\approx 71.9^\circ$

2. Prove that if $a \parallel b$ and $b \parallel c$ in space then $a \parallel c$

ANSWER Transitivity of direction

3. Give two skew lines that are perpendicular in a cube

ANSWER AB and B_1C_1

4. How many pairs of skew edges has a cube?

ANSWER 24

5. Prove a line parallel to two intersecting planes is parallel to their line of intersection

ANSWER It is parallel to a line in each; the common direction is the intersection

6. Find the common perpendicular of AB and CC_1 in a unit cube

ANSWER BC , length 1

7. Explain why the angle between skew lines does not depend on where you translate

ANSWER Translation preserves direction

07 Homework

Homework — 5 tasks

Draw the cube once and answer tasks 1–3 from that one drawing.

- For cube $ABCD A_1 B_1 C_1 D_1$, list the edges meeting, parallel to and skew to CD .
- Find the angle between AD and BC_1 .
- Find the angle between the face diagonals AC and BD_1 .
- Give an example showing two planes parallel to the same line need not be parallel.
- Explain in three sentences the difference between parallel and skew.